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BAGONG PILIPINAS

CC 101 Laboratory Activity: Exploring the Windows Command Line

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Instructor: Mr. John Mark D. Gabrentina

Introduction (3 to 5 sentences): What is the command line and the purpose of this activity, in your own words.

- The command line is basically a screen where you type commands to control your computer instead of clicking on icons or buttons with a mouse. We did this lab to learn how to use Command Prompt and PowerShell to check system info, create folders, and manage files without using the regular desktop. It helps us understand how operating systems work behind the scenes, so we get comfortable with tech tools we'll need for computer science

Item	Details
Course	CC 101 Introduction to Computing (BSCS 1st Year)
Course Outcome	CO2: Identify the functions and relationships of major hardware and software components
Output	Command Line Exploration Documentation (PDF)
Estimated time	2 to 2½ hours
Reference	CC 101 Lecture Notes: Exploring the Windows Command Line

Instructions

Follow every step in order, take the screenshots marked **[Screenshot]**, and answer the questions marked **[Answer]** in your documentation.

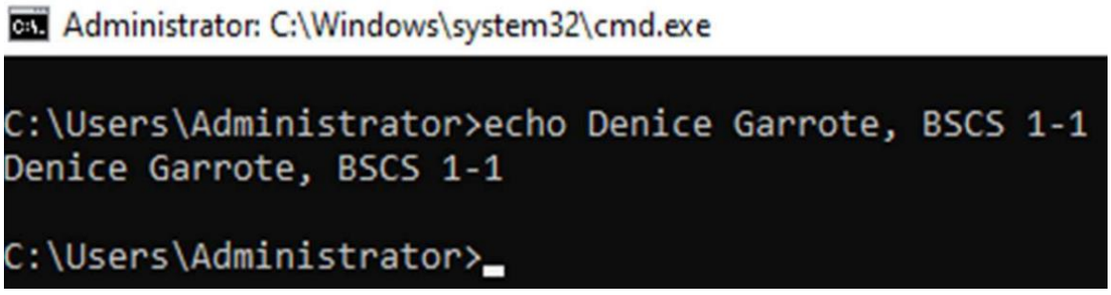
Before you start

- You need a Windows 10 or 11 computer. No administrator rights are needed.
- To take a screenshot, press **Windows + Shift + S**, drag over the terminal window, then paste it (Ctrl + V) into your documentation.
- Replace the sample values (Juan Dela Cruz, BSCS 1-1) with **your own** details.
- Type every command yourself. Do not copy and paste; typing is how you learn.

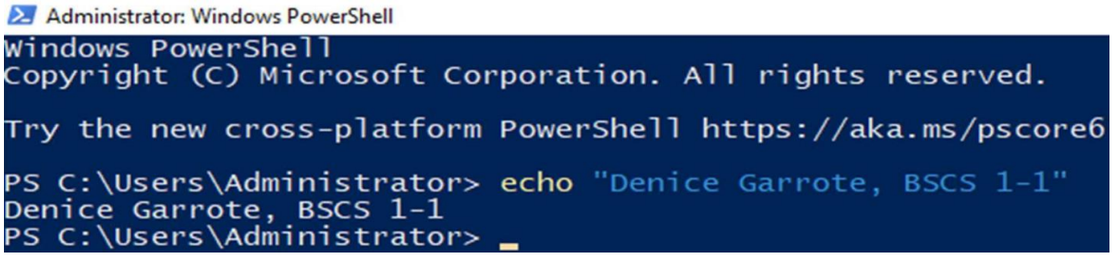
Part A: Open the Two Shells

1. Press Windows + R, type cmd, and press Enter. Type cls and press Enter.
2. Type echo Juan Dela Cruz, BSCS 1-1 (your own name and section) and press Enter.

[Screenshot] Figure A1



3. Press Windows + R, type PowerShell, and press Enter. Type the same echo command but put the text in quotation marks: echo "Juan Dela Cruz, BSCS 1-1". PowerShell treats the comma as a list separator without them. **[Screenshot] Figure A2**



[Answer] A-Q1. Compare the two prompts. What is different, and what does each part of the Command Prompt mean?

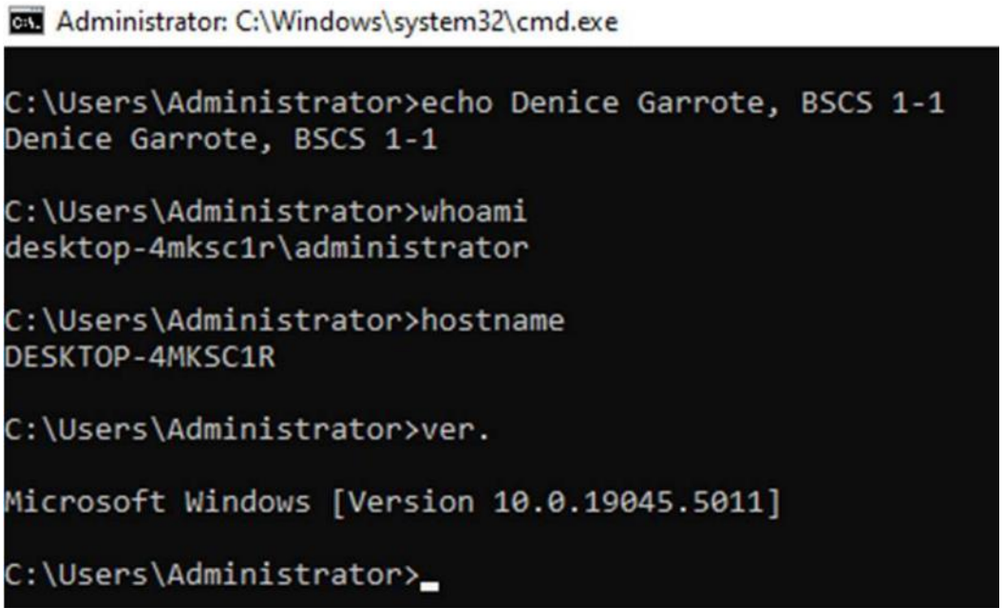
- Figure A1 shows the old-school Command Prompt, which jumps straight into the folder path `C:\Users\Administrator>` and prints text as you type it. PowerShell in Figure A2 is a bit more modern, tagging PS onto the front of the path and needing quotes around text with commas, so it doesn't get confused. As for the path itself, `C:` is the main hard drive, `Users\Administrator` is the exact user folder you're working in, and the final `>` symbol simply means the system is waiting for your next command.

Part B: Inspect Your System (Command Prompt)

Use the Command Prompt window for Parts B to D.

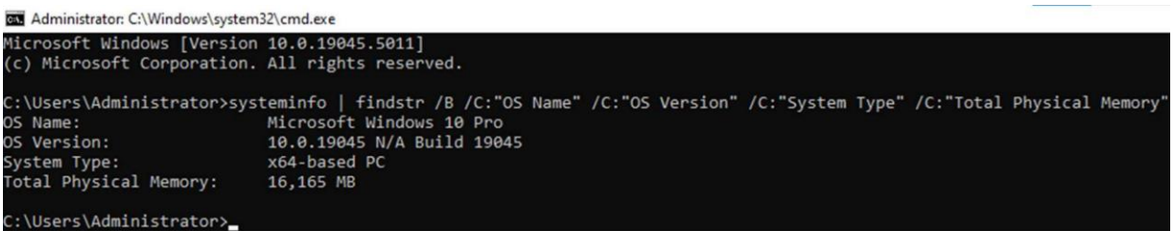
1. Run these commands one at a time: whoami, then hostname, then ver. **[Screenshot]**

Figure B1

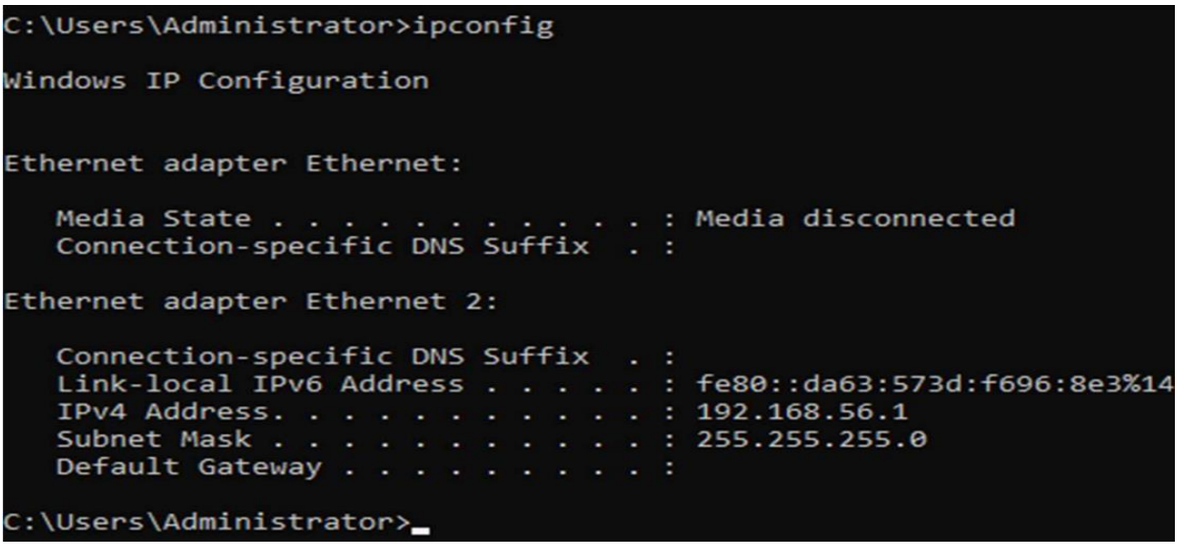


2. Run the command below. It may take 10 to 30 seconds. **[Screenshot] Figure B2**

```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version" /C:"System Type" /C:"Total
Physical Memory"
```



3. Run ipconfig. Find the adapter that shows an IPv4 Address. **[Screenshot] Figure B3**



[Answer] B-Q1. Complete this table in your documentation using your results:

Item	Your result
Logged-in account (whoami)	desktop-4mksc1r\administrator
Computer name (hostname)	DESKTOP-4MKSC1R
OS name	Microsoft Windows 10 pro
OS version	10.0.19045 N/A Build 19045
System type (x64 or ARM)	x64-based PC
Total physical memory (RAM)	16,165 MB
IPv4 address	192.168.56.1

[Answer] B-Q2. The | symbol in step 2 is called a **pipe**. Run systeminfo alone and compare. In your own words, what did | findstr do?

- Running `systeminfo` by itself dumps a massive wall of system details all over your screen. The pipe (|) takes all that output and hands it right to `findstr`, which acts like a search filter. Instead of making you scroll through everything, it grabs and displays only the exact lines you asked for, like OS version or total RAM.

Part C: Build a Folder Structure

1. Go to your user folder and create the lab folders:

```
cd %USERPROFILE%
mkdir CC101-Lab
cd CC101-Lab
mkdir Notes Projects Screenshots
```

2. Create a text file about yourself (use your own details):

```
echo Name: Juan Dela Cruz > profile.txt
echo Section: BSCS 1-1 >> profile.txt
echo Favorite app: YouTube >> profile.txt
type profile.txt
```

[Screenshot] Figure C1 (show all commands and the output of type)

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.5011]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd %USERPROFILE%

C:\Users\Administrator>mkdir CC101-Lab
A subdirectory or file CC101-Lab already exists.

C:\Users\Administrator>cd CC101-Lab

C:\Users\Administrator\CC101-Lab>mkdir Notes Projects Screenshots
A subdirectory or file Notes already exists.
Error occurred while processing: Notes.
A subdirectory or file Projects already exists.
Error occurred while processing: Projects.
A subdirectory or file Screenshots already exists.
Error occurred while processing: Screenshots.

C:\Users\Administrator\CC101-Lab>echo Name: Denice Garrote > profile.txt

C:\Users\Administrator\CC101-Lab>echo Section: BSCS 1-1 >> profile.txt

C:\Users\Administrator\CC101-Lab>echo Favorite app: Instagram >> profile.txt

C:\Users\Administrator\CC101-Lab>type profile.txt
Name: Denice Garrote
Section: BSCS 1-1
Favorite app: Instagram

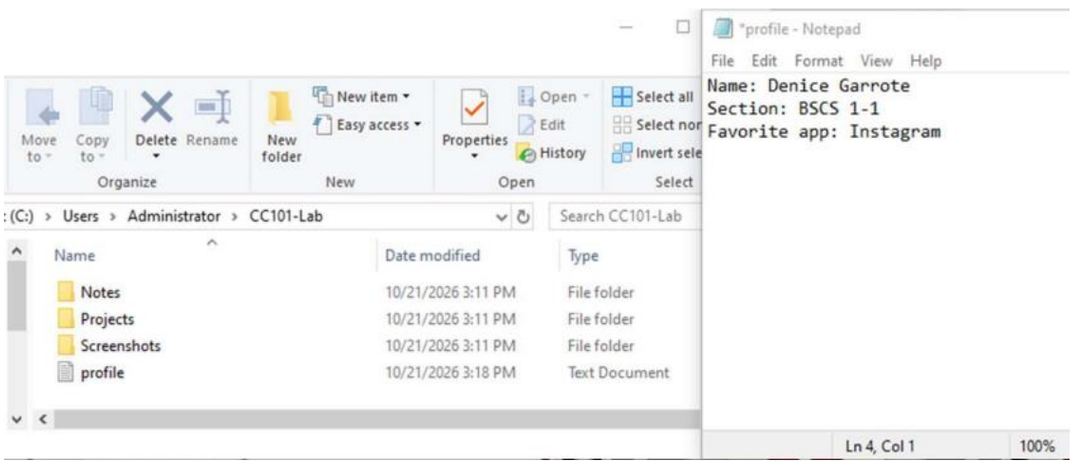
C:\Users\Administrator\CC101-Lab>
```

3. Run tree /F to display your folder structure. **[Screenshot] Figure C2**

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.5011]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>tree/F CC101-Lab
Folder PATH listing
Volume serial number is 00000076 2862:AB55
C:\USERS\ADMINISTRATOR\CC101-LAB
|   profile.txt
|
+---Notes
+---Projects
\---Screenshots
```

4. Open File Explorer, go to C:\Users\(your account)\CC101-Lab, and open profile.txt in Notepad. **[Screenshot] Figure C3** (File Explorer and Notepad side by side)



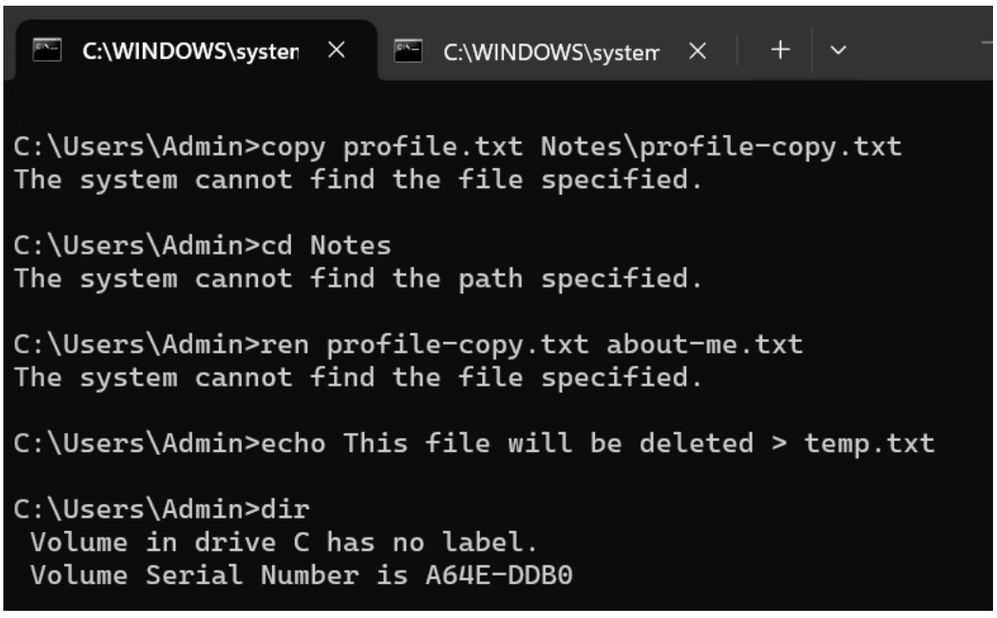
[Answer] C-Q1. What would happen if you used > instead of >> in the second echo line? Test it on a new file (test.txt) and explain.

- Using a single > completely wipes out whatever was already in `test.txt` and replaces it with your new line, so you're left with just `line 2`. If you want to keep the original content and simply tack `line 2` onto the end, you need to use >> instead.

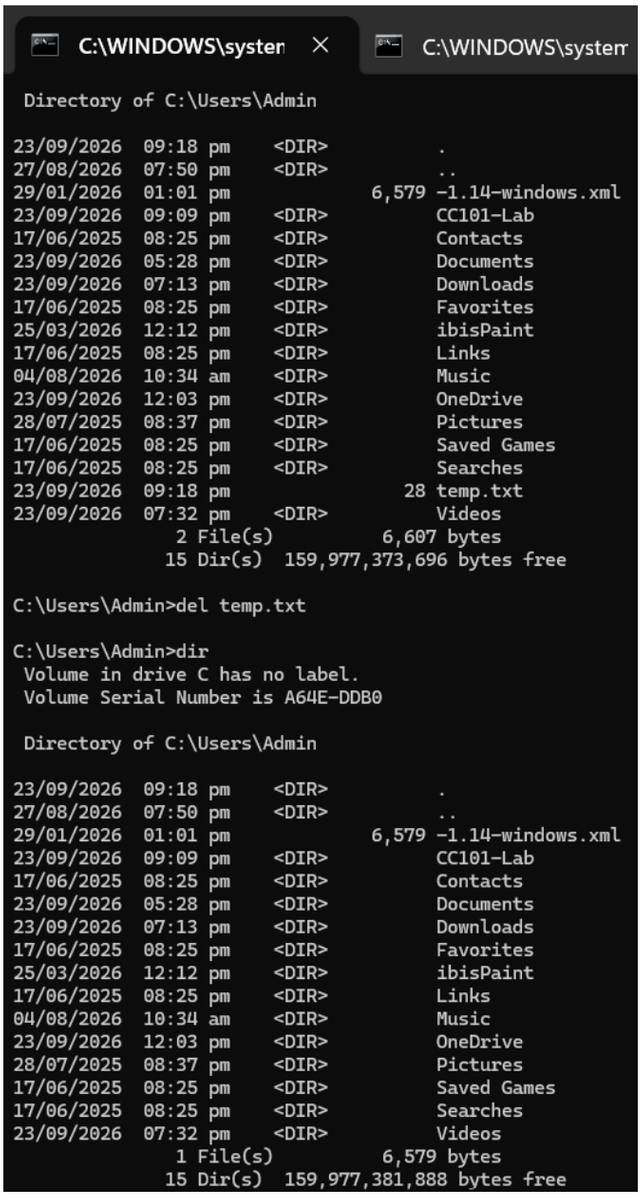
Part D: Copy, Rename, and Delete Files

```
copy profile.txt Notes\profile-copy.txt
cd Notes
ren profile-copy.txt about-me.txt
echo This file will be deleted > temp.txt
dir
del temp.txt
dir
cd ..
tree /F
```

[Screenshot] Figure D1 (from copy to the first dir)



and **[Screenshot] Figure D2** (from del to tree /F)



[Answer] D-Q1. After del temp.txt, check your Recycle Bin. Is temp.txt there? What does this tell you about deleting files from the command line?

- No, `temp.txt` will **not** be in the Recycle Bin. Deleting a file from the command line using `del` bypasses the Recycle Bin entirely and permanently removes the file's index entry from the file system immediately. This means command-line deletions are direct and non-recoverable through standard Windows to undo mechanisms, so you must be extra careful when running to delete commands in the terminal.

Part E: Explore PowerShell

Switch to the PowerShell window.

1. Go to your lab folder and list and read its files:

```
cd $HOME\CC101-Lab
Get-ChildItem
Get-Content profile.txt
$PSVersionTable
```

[Screenshot] Figure E1

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Admin> cd $HOME\CC101-Lab
PS C:\Users\Admin\CC101-Lab> Get-ChildItem


    Directory: C:\Users\Admin\CC101-Lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        23/09/2026   9:36 pm                Notes
d-----        23/09/2026   9:36 pm                Projects
d-----        23/09/2026   9:36 pm                Screenshots
-a----        23/09/2026   9:37 pm             69 profile.txt


PS C:\Users\Admin\CC101-Lab> Get-Content profile.txt
Name: Denice Garrote
Section: BSCS 1-1
Favorite app: Instagram
PS C:\Users\Admin\CC101-Lab> $PSVersionTable

Name                           Value
----                           -----
PSVersion                      5.1.26100.9444
PSEdition                      Desktop
PSCompatibleVersions           {1.0, 2.0, 3.0, 4.0...}
BuildVersion                   10.0.26100.9444
CLRVersion                     4.0.30319.42000
WSManStackVersion              3.0
PSRemotingProtocolVersion      2.3
SerializationVersion           1.1.0.1


PS C:\Users\Admin\CC101-Lab>
```

2. Run Get-Help Get-ChildItem. If PowerShell asks whether to run Update-Help, type N and press Enter. **[Screenshot] Figure E2**

```
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Admin> Get-Help Get-ChildItem

NAME
    Get-ChildItem

SYNTAX
    Get-ChildItem [[-Path] <string[]>] [[-Filter] <string>]
    [-Include <string[]>] [-Exclude <string[]>] [-Recurse]
    [-Depth <uint32>] [-Force] [-Name] [-UseTransaction]
    [-Attributes {ReadOnly | Hidden | System | Directory |
    Archive | Device | Normal | Temporary | SparseFile |
    ReparsePoint | Compressed | Offline | NotContentIndexed |
    Encrypted | IntegrityStream | NoScrubData}]
    [-FollowSymlink] [-Directory] [-File] [-Hidden]
    [-ReadOnly] [-System]  [<CommonParameters>]

    Get-ChildItem [[-Filter] <string>] -LiteralPath <string[]>
    [-Include <string[]>] [-Exclude <string[]>] [-Recurse]
    [-Depth <uint32>] [-Force] [-Name] [-UseTransaction]
    [-Attributes {ReadOnly | Hidden | System | Directory |
    Archive | Device | Normal | Temporary | SparseFile |
    ReparsePoint | Compressed | Offline | NotContentIndexed |
    Encrypted | IntegrityStream | NoScrubData}]
    [-FollowSymlink] [-Directory] [-File] [-Hidden]
    [-ReadOnly] [-System]  [<CommonParameters>]


ALIASES
    gci
    ls
    dir


REMARKS
    Get-Help cannot find the Help files for this cmdlet on
    this computer. It is displaying only partial help.
        -- To download and install Help files for the module
    that includes this cmdlet, use Update-Help.
        -- To view the Help topic for this cmdlet online,
    type: "Get-Help Get-ChildItem -Online" or
          go to
    https://go.microsoft.com/fwlink/?LinkID=113308.

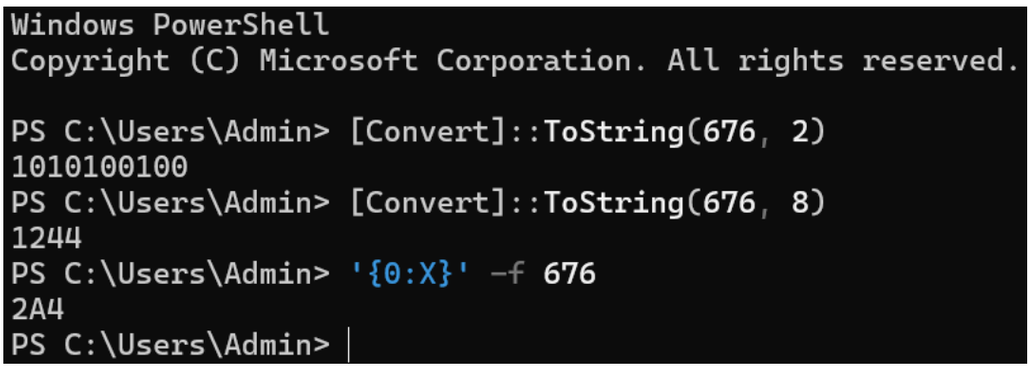

PS C:\Users\Admin>
```

3. Take the **last three digits of your student ID number** (for example, 2025-00347 →

347). Convert it with PowerShell:

```
[Convert]::ToString(347, 2)
[Convert]::ToString(347, 8)
'{0:X}' -f 347
```

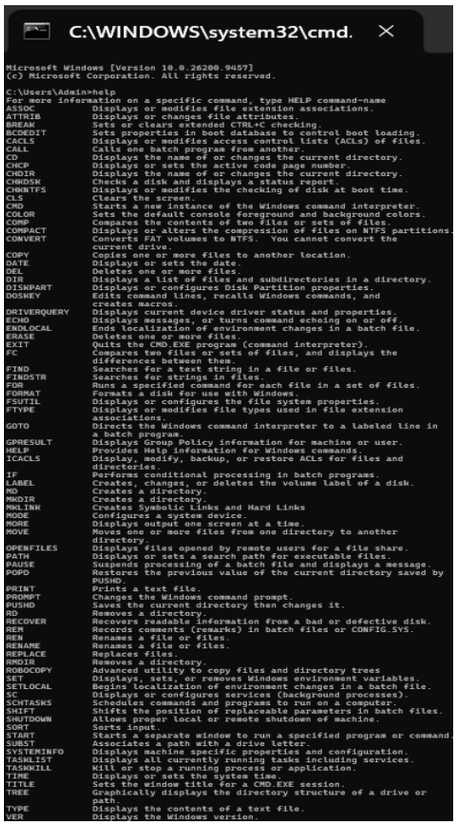
[Screenshot] Figure E3



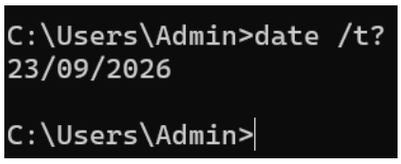
[Answer] E-Q1. Write your number in decimal, binary, octal, and hexadecimal. Keep this; you will verify it by hand in Week 5.

Part F: Discover a New Command

1. In Command Prompt, type help to see a list of commands.
2. Choose **one** command that was **not** taught in this lesson and is safe to run (for example: date /t, time /t, tasklist, vol, or color 0A).
3. Read its help with command /? (for example, tasklist /?), then run it. **[Screenshot] Figure F1** (help output)



and **[Screenshot] Figure F2** (command result)



[Answer] F-Q1. What does your chosen command do, and when would it be useful?

- The command '`date /t`' displays the current system date without prompting to enter a new date.

- It is useful in scripts, batch files, or quick terminal checks when you simply need to display or log the current date without accidentally changing the system time.

Part G: Reflection

[Answer] Answer in 3 to 5 sentences each:

1. **G-Q1.** Which task in this activity was faster in the CLI than in the GUI? Which was easier in the GUI? Explain why.
   - Creating multiple subfolders at the same time using `mkdir Notes Projects Screenshots` was much faster in the command line because I could do it all in one single line instead of right-clicking and making folders one by one. On the other hand, opening a file and viewing its text side-by-side felt a lot easier in the GUI. With the GUI, I could just double-click `profile.txt` to open Notepad and snap windows around with my mouse, which felt more natural than using text commands.
2. **G-Q2.** What error did you encounter (if any), what caused it, and how did you fix it?
   - I ran into an error when I typed `mkdir CC101-Lab` and `mkdir Notes Projects Screenshots` because those folders were already created on my computer from a previous run. The terminal displayed messages saying "A subdirectory or file CC101-Lab already exists" and "Error occurred while processing: Notes". I fixed it simply by using `cd CC101-Lab` to move into the existing folder instead of trying to create it again, and then I continued with my remaining commands.
3. **G-Q3.** Give one real-world job or situation in computing where knowing the command line would be important.
   - Knowing the command line is important for IT to support specialists or system administrators who need to fix computers remotely. When a computer's graphical interface freezes, fails to load, or isn't accessible, tech support can still log in through a remote terminal to check system specs, fix file permissions, or run network diagnostics. Without command-line skills, it would be almost impossible to repair system issues in those situations.

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Documentation and Submission

Submit one PDF that records what you did, what the computer showed, and what you learned. Someone reading it should be able to repeat your work.

Required Contents (in this order)

1. **Cover page:** activity title, full name, student ID, section, date performed, and instructor's name.
2. **Introduction** (3 to 5 sentences): what the command line is and the purpose of this activity, in your own words.
3. **Procedure and results, Parts A to F:** for each part, a short heading, the commands you typed, the labeled screenshots, and your answers to that part's [Answer] questions.
4. **Commands learned table:** every command you used and what it does, in your own words.
5. **Reflection:** your answers to G-Q1 to G-Q3.

Formatting Rules

Item	Requirement
File type	PDF (write in Word or Google Docs, then export as PDF)
File name	CC101_CLI-Lab_Lastname_Firstname.pdf
Font	Arial or Calibri, 11 pt; commands in Consolas or Courier New
Screenshots	Crop to the terminal window; text must be readable. Label each one below the image, e.g., "Figure B2. Filtered system info output"
Personal details	Your own name, section, and student ID must appear in Figures A1, C1, and E3
Total screenshots	16 (A1–A2, B1–B3, C1–C3, D1–D2, E1–E3, F1–F2)

Sample Commands Learned Table

Command	Shell	What it does
mkdir Notes	Command Prompt	Creates a folder named Notes in the current folder
type profile.txt	Command Prompt	Displays the contents of profile.txt

Submission

- Upload the PDF to the activity page in the course LMS before the deadline your instructor sets.
- Outputs must be your own work. Identical screenshots or answers will be treated as plagiarism under the class policy.
- Keep your CC101-Lab folder; later activities may build on it.

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Grading Rubric

The documentation is scored out of 100 points and counts under Laboratory: Project.

Criterion	Points	Full marks	Partial marks	Low marks
Completeness of procedure	30	All 16 labeled, readable screenshots; every part A to F documented	1 to 4 screenshots missing or unreadable	5 or more screenshots missing
Accuracy of results	20	System-information table complete; commands and conversions correct and personalized	Minor errors or 1 to 2 missing values	Many errors, or results not the student's own
Answers to guide questions (A to F)	20	Correct, clear, in the student's own words	Mostly correct but brief or vague	Missing, incorrect, or copied
Reflection (G-Q1 to G-Q3)	15	Specific, honest, and linked to the activity	General statements with few specifics	Missing or off-topic
Organization and formatting	15	Follows the required order, naming, and formatting rules; commands learned table complete	Some rules not followed	Disorganized; wrong file type or name
Total	**100**			